

HOMELAB NETWORK SECURITY SERVICES AUDITING

Lab 2 - Local Home Lab Auditing

Assessment Environment: Isolated VirtualBox Lab

Assessment Network: 10.10.10.0/27

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1. Executive Summary

This assessment evaluated a small isolated network containing a Kali Linux audit workstation, a Windows 7 client and an Ubuntu Server. The assessment focused on network discovery, service enumeration, protocol security, vulnerability identification and basic remediation verification.

The most significant issue identified was the Windows host's exposure to SMBv1 and the associated MS17-010/CVE-2017-0143 vulnerability. Before remediation, Nmap confirmed SMBv1 support and reported the host as vulnerable to MS17-010. SMBv1 was subsequently disabled, and a follow-up MS17-010 check no longer reported the vulnerability.

The Ubuntu server exposed SSH and HTTP. Apache was serving the default Ubuntu page, while the host firewall was initially inactive. These conditions were documented as configuration/exposure observations rather than as confirmed software vulnerabilities. The assessment did not identify a confirmed vulnerability in the Apache service from the performed Nmap checks.

2. Scope and Environment

Asset	Address	Role
Kali Linux	10.10.10.10	Security assessment workstation
Windows 7	10.10.10.20	Client / assessment target
Ubuntu Server	10.10.10.30	Linux server / assessment target
VirtualBox host interface	10.10.10.1	Host-side network interface

The assessment was performed against the isolated 10.10.10.0/27 laboratory network. Testing was limited to the systems listed above.

3. Methodology

The assessment followed a simplified professional network-security-audit workflow:

Phase	Activities
Discovery	ICMP/ARP-based host discovery using Nmap
Enumeration	Full TCP-port scanning, service/version detection and targeted NSE scripts
UDP assessment	Top-100 UDP port scan
Protocol assessment	SMB protocol and SMB security-mode enumeration
Vulnerability assessment	Nmap vulnerability NSE scripts against lab targets
Configuration review	Windows firewall/network state and Ubuntu listening services/firewall state
Remediation verification	Repeated targeted Nmap checks after Windows SMB remediation

4. Asset and Service Inventory

Host	Observed services
10.10.10.20 / Windows	135/tcp MSRPC; 139/tcp NetBIOS; 445/tcp SMB; 49153-49156/tcp MSRPC; 137/udp NetBIOS
10.10.10.30 / Ubuntu	22/tcp OpenSSH 10.2p1; 80/tcp Apache 2.4.66

The Windows host exposed SMB-related services and dynamic RPC ports. The Ubuntu host exposed SSH and HTTP. The UDP scan identified UDP/137 on the Windows host and no open top-100 UDP ports on Ubuntu.

5. Findings Summary

ID	NET-	Finding	Host	Severity	Status
001		SMBv1 / MS17-010 exposure	10.10.10.20	Critical	Remediated in lab
NET-002		SMB message signing disabled	10.10.10.20	High	Observed; remediation not verified
NET-003		SMBv1 protocol enabled	10.10.10.20	High	Remediated in lab
NET-004		Default Apache HTTP service exposed	10.10.10.30	Low	Configuration observation

6. Detailed Findings

NET-001 — SMBv1 / MS17-010 Remote Code Execution Exposure

Affected host: 10.10.10.20 ; **Severity:**Critical ; **Status:** Remediated in lab

The initial Nmap vulnerability assessment reported the Windows host as vulnerable to MS17-010, associated with CVE-2017-0143. The vulnerability is associated with the legacy SMBv1 protocol.

Evidence: scans/before/windows-vuln.txt reported smb-vuln-ms17-010 as VULNERABLE. The initial SMB protocol scan also reported NT LM 0.12 (SMBv1).

Impact: A vulnerable SMB service can expose a Windows system to remote code execution attacks from an attacker with suitable network access.

Remediation performed: SMBv1 was disabled through the Windows registry. A follow-up SMB protocol scan no longer reported SMBv1, and the targeted MS17-010 NSE check did not report the vulnerability. **Verification:** scans/after/windows-smb.txt contains only SMBv2 dialects 2.0.2 and 2.1. scans/after/ms17-010.txt did not report the vulnerability.

NET-002 — SMB Message Signing Disabled

Affected host:10.10.10.20 ;**Severity:** High ; **Status:** Observed; after-state not verified

The initial SMB security assessment reported message signing as disabled. This is a security-hardening weakness because SMB signing can provide integrity protection for SMB communications and can reduce exposure to certain relay or man-in-the-middle attack scenarios.

Evidence: scans/windows-smb.txt reported message_signing: disabled.

Recommendation: Enable SMB signing through appropriate Windows security policy or registry configuration where operationally appropriate, and verify the resulting state with a compatible SMB security assessment.

Reporting limitation: The post-remediation smb-security-mode scan returned no signing state, so this report does not claim that SMB signing was successfully remediated.

NET-003 — SMBv1 Protocol Enabled

Affected host:1 0.10.10.20 ; **Severity:** High ; **Status:** Remediated in the laboratory

The initial SMB protocol enumeration identified the legacy SMBv1 dialect (NT LM 0.12). SMBv1 is obsolete and has a significant security history, including its relationship to MS17-010.

Remediation: SMBv1 was disabled on the Windows system.

Verification: The post-remediation SMB protocol scan no longer listed SMBv1 and reported only SMB 2.0.2 and SMB 2.1 dialects.

NET-004 — Default Apache HTTP Service Exposed

Affected host: 10.10.10.30 ; **Severity:** Low ; **Status:** Configuration observation

The Ubuntu server was listening on TCP/80 and Nmap identified Apache HTTP Server 2.4.66. HTTP enumeration showed the default Apache2 Ubuntu page. The local Ubuntu review also showed Apache listening on all interfaces and UFW initially inactive.

Evidence: Nmap identified TCP/80 as open and reported the Apache default page. `ss -tulpen` showed Apache listening on port 80. `ufw status verbose` initially reported Status: inactive.

Impact: The default page is not itself evidence of a critical vulnerability, but unnecessary HTTP exposure and default configuration can increase attack surface and disclose implementation details. **Recommendation:** If HTTP is required, keep the service but restrict access through firewall/network policy as appropriate, replace the default site with the intended application, and harden the web server. If HTTP is not required, disable it. The appropriate action depends on the actual business requirement.

7. Before / After Verification

Control	Before	After	Assessment
SMBv1	Detected	Not detected	Remediated
MS17-010	VULNERABLE	Not reported by targeted NSE check	Remediated in lab; patch state should still be verified
SMB signing	Disabled	Not reported	Not verified
Ubuntu HTTP	TCP/80 open	No final remediation scan supplied	Observation retained
Ubuntu firewall	Inactive	No final state supplied	No remediation claim

The Windows before/after evidence demonstrates a measurable change in the SMB attack surface. The report deliberately distinguishes confirmed remediation from controls whose post-remediation state was not captured.

8. Risk Assessment

Risk ratings in this report are qualitative and based on the observed network exposure, exploitability and potential security impact within the laboratory context. They are not CVSS scores.

Finding	Primary risk rationale
NET-001	Legacy SMB exposure combined with a confirmed MS17-010 detection creates a direct high-impact remote attack surface.
NET-002	Disabled SMB signing weakens SMB communication integrity and can increase exposure to relay/MITM scenarios depending on configuration.
NET-003	SMBv1 is an obsolete protocol and expands the attack surface; it also directly relates to the MS17-010 exposure observed in the lab.
NET-004	Unnecessary/default HTTP exposure increases attack surface but no specific Apache vulnerability was confirmed by the performed tests.

9. Remediation Recommendations

1. Remove SMBv1 from all Windows systems where it is not required.
2. Keep Windows systems fully patched and replace unsupported operating systems with supported versions.
3. Enable SMB signing where appropriate and verify the configuration after deployment.
4. Restrict SMB (TCP/139 and TCP/445) to hosts and network segments that require it.

5. Review the business requirement for Apache/HTTP and restrict access with host/network firewall rules where appropriate.
6. Replace default web-server content with intended production content and remove unnecessary information disclosure.
7. Maintain a repeatable audit-and-retest process so every security finding has evidence before remediation and evidence after remediation.

10. Limitations

The assessment was performed in an isolated virtual laboratory rather than a production network. Nmap results identify observable network exposure and provide useful indicators, but they do not replace authenticated configuration review, vendor patch verification, or a complete vulnerability-management process.

The Windows SMB signing state was not confirmed after remediation because the post-remediation Nmap script returned no signing-state output. No claim of successful SMB-signing remediation is therefore made.

The Ubuntu firewall was initially inactive, but no final firewall state was supplied for this report. Accordingly, the report does not claim that Ubuntu firewall remediation was completed.

11. Conclusion

The assessment demonstrated a complete practical network-security-audit workflow: host discovery, service enumeration, protocol assessment, vulnerability identification, evidence collection and remediation verification. The most significant laboratory exposure was the Windows SMBv1/MS17-010 issue, which was successfully reduced by disabling SMBv1 and verified through repeated Nmap assessment.

The project provides a reusable foundation for future audit engagements: establish scope, collect raw evidence, identify and justify findings, remediate where appropriate, and perform a repeatable re-test rather than relying only on a one-time scan.

Appendix A — Evidence Files

network-security-audit/README.md
network-security-audit/findings/findings.md
network-security-audit/remediation/remediation.md
network-security-audit/scans/after/ms17-010.txt
network-security-audit/scans/after/windows-smb.txt
network-security-audit/scans/before/windows-smb.txt
network-security-audit/scans/before/windows-vuln.txt
network-security-audit/scans/discovery.txt
network-security-audit/scans/tcp-all-ports.txt
network-security-audit/scans/tcp-services-ubuntu-server.txt
network-security-audit/scans/ubuntu-services-http-ssh.txt
network-security-audit/scans/udp-services.txt
network-security-audit/scans/vulnerability-ubuntu-server.txt
network-security-audit/scans/vulnerability-windows.txt
network-security-audit/scans/windows-services.scan.txt
network-security-audit/scans/windows-smb.txt